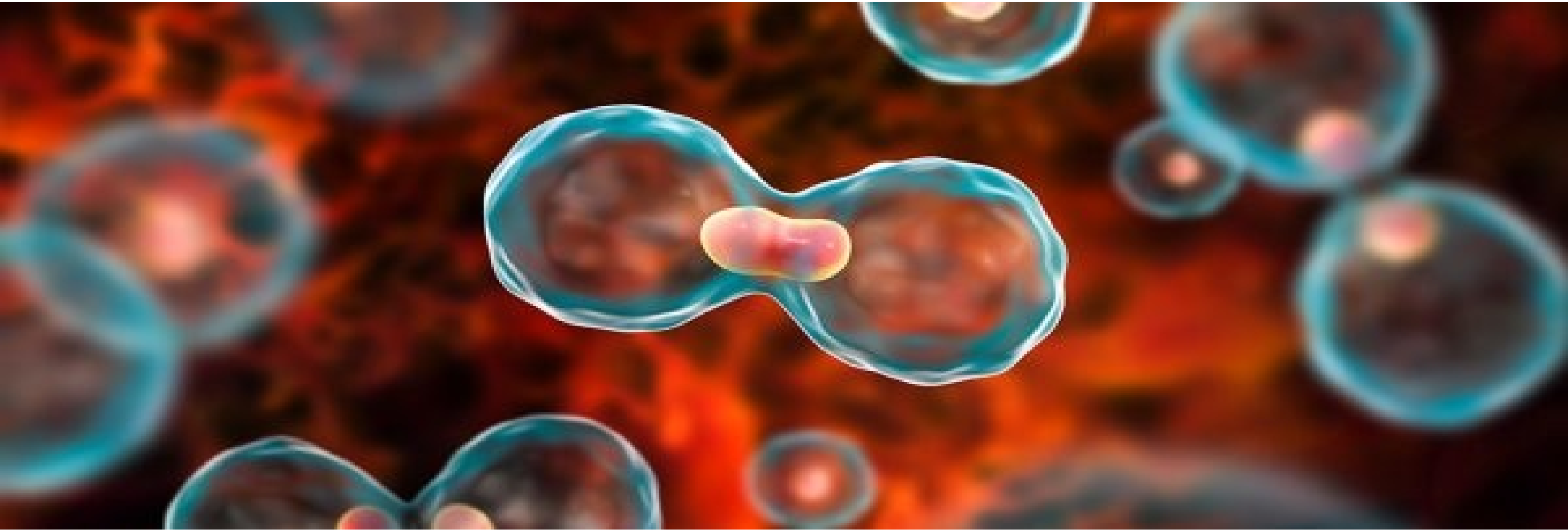
Title of the lecture:

Biochemistry of cancer



Professor

Dr. Salar A. Ahmed

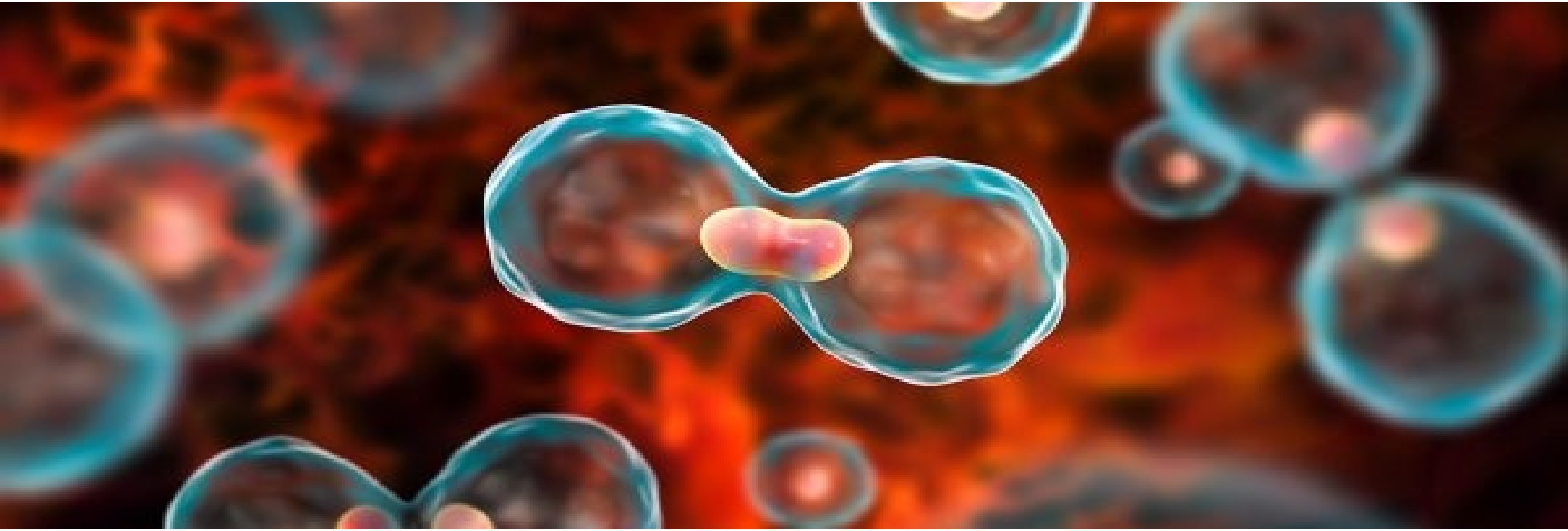
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Title of the lecture:

Biochemistry of cancer



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Aims & objectives

- **Definition of cancer & metastasis.**
- **Differences between benign & malignant neoplasms.**
- **Role of biochemistry in cancer.**
- **To describe the causes, chemical carcinogens, & chemotherapeutic agents used in the treatment.**
- **To understand how indirect carcinogens are converted into direct carcinogens.**
- **To understand cancer converted biochemical changes**

What is a Neoplasm (Tumor)

It is an area of tissue whose growth has become independent of the surrounding tissues.

What are the different types of neoplasms?

Classification of Neoplasms: Benign vs. Malignant

Malignant neoplasm refers to Malignant growth or carcinoma (pre-malignant) = cancer or malignant tumor.

- **Benign neoplasm refers to a benign growth or tumor.**

What are the characteristics of benign & malignant cells?

Benign neoplasm	Malignant neoplasm
Grow Slowly	Multiply rapidly
The cells are normal	Cells are abnormal and immortal
Does not spread to distant sites(non metastatic), Remain localized	Metastatic, spread to other parts of the body & Invasion of local tissues
Cell growth is balanced with cell death	Proliferation exceeds the death rate
Only kills if damaging vital organs	Always kills, if it is not diagnosed early and treated

What is cancer?

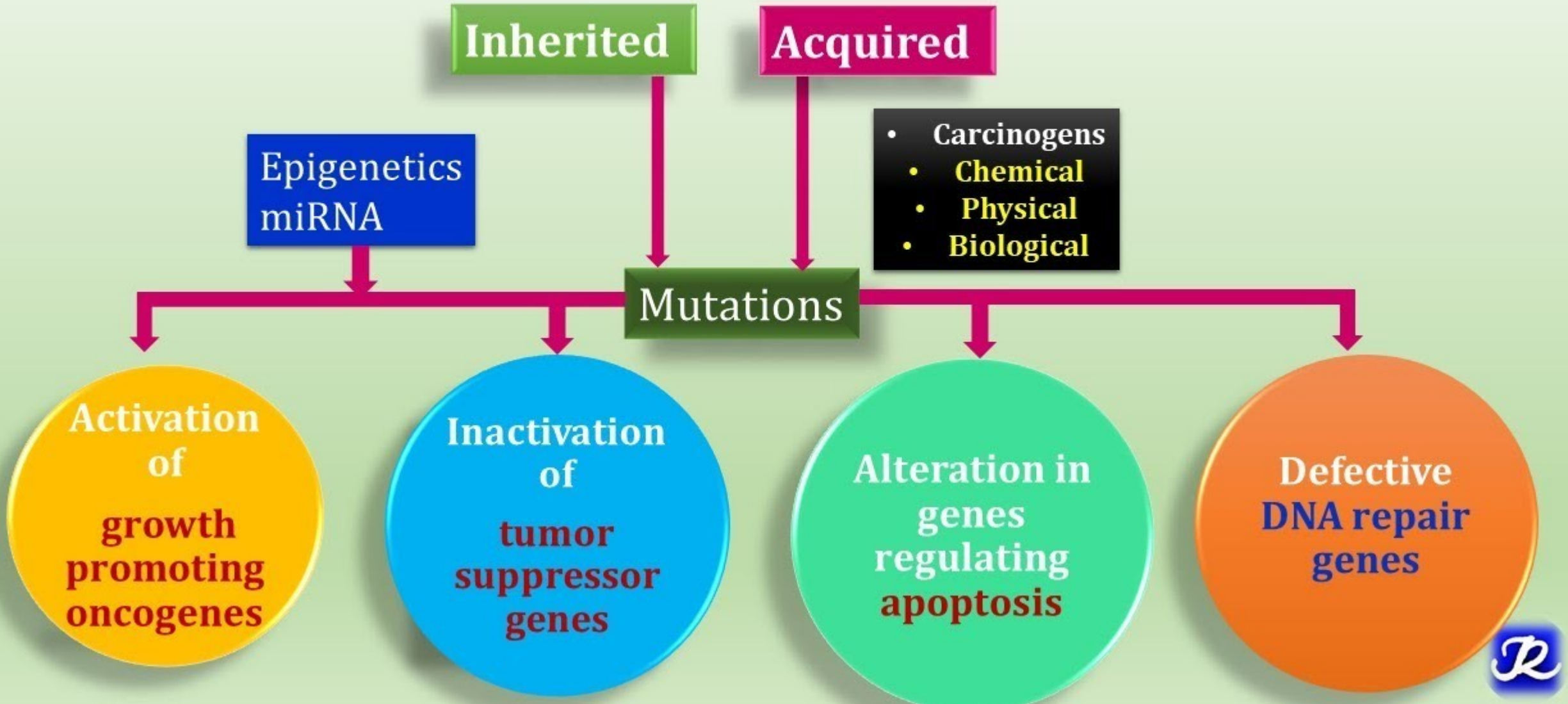
It is a group of diseases involving abnormal cell growth with the potential to invade or spread to other parts of the body.

Is cancer a genetic disease?

- **It is a long-term multistage genetic process.**
- **The first stage is when some form of carcinogens damages the DNA**
- **These harmful agents can alter or damage DNA, as well as associated proteins and enzymes.**

MOLECULAR BASIS OF CANCER

Cancer/06

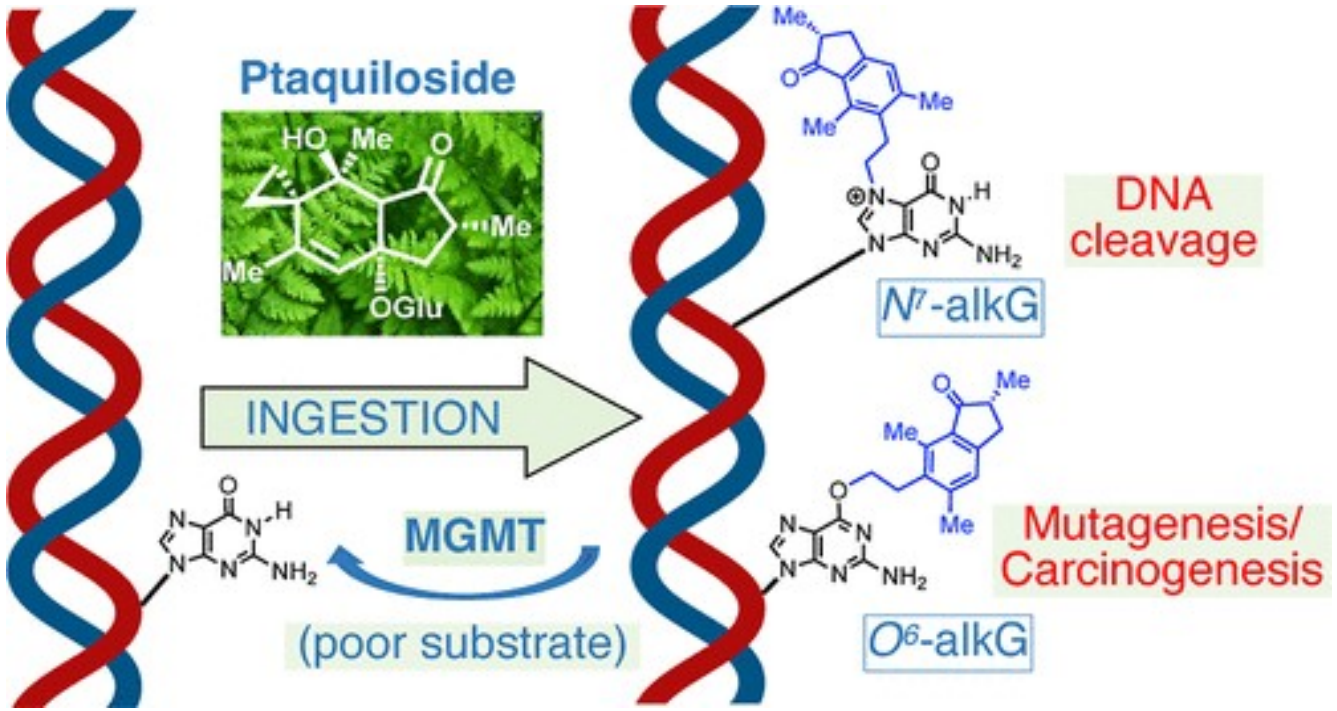


How do carcinogens interact with DNA to cause genetic damage?

DNA damage can be caused by Covalent binding:

- **Carcinogens form covalent bonds with cellular components, including DNA, RNA, & proteins.**
- **They primarily target the purine, pyrimidine, or phosphodiester groups in DNA, with guanine (G) being the most frequent site of interaction.**
- **These covalent attachments can cause various types of DNA damage. While cellular repair mechanisms can fix some of this damage, any unrepaired lesions may result in mutations.**

- **Additional damage occurs, eventually leading to chromosome breakdown and rearrangement.**
- **This process produces a new phenotype that loses control over the process of mitosis.**



How are malignant neoplasms classified?

Classification of malignant neoplasms

1-The carcinomas:

The carcinomas are solid tumors derived from epithelial tissues that line external and internal body surfaces, including: Skin, Glands, Digestive, Urinary, and Reproductive organs

Such as:

- **Stomach (gastric adenocarcinoma)**
- **Colon (adenocarcinoma of the colon)**
- **Liver (hepatocellular carcinoma or hepatoma)**

2. The sarcomas:

Derived from connective tissues such as

- **Bone,**
- **Muscle,**
- **Cartilage,**
- **Bone marrow, and**
- **Cells of the lymphatic system**

Such as:

- **Bone (Osteosarcoma),**
- **Plasma cells (Multiple myeloma)**
- **Leukemias & Lymphomas**

3- Mixed tissue tumors

Derived from tissue that is capable of differentiation into both epithelial and connective tissue,

Examples

Like Teratoma: It arises from germ cells that can develop into eggs or sperm

What are the most common causes of cancer?

Causes of malignancy

Many, if not all, likely have multifactorial causes.

The principal causes of malignant neoplasms are the following :

1- Predisposing factors (20%) such as genetic, age, sex, race, & obesity.

2. Environmental factors, such as:

- **Radiation (Radiant energy).**
- **Chemicals**
- **Viruses**
- **Hormones**
- **Immunological**

How does radiation (Radiant energy (Physical)) cause cancer?

1st cause of cancer: radiation

A- Radiation, whatever its source is a wave of energy. When this energy interacts with DNA, it causes DNA damage and mutations that lead to cancer

Examples:

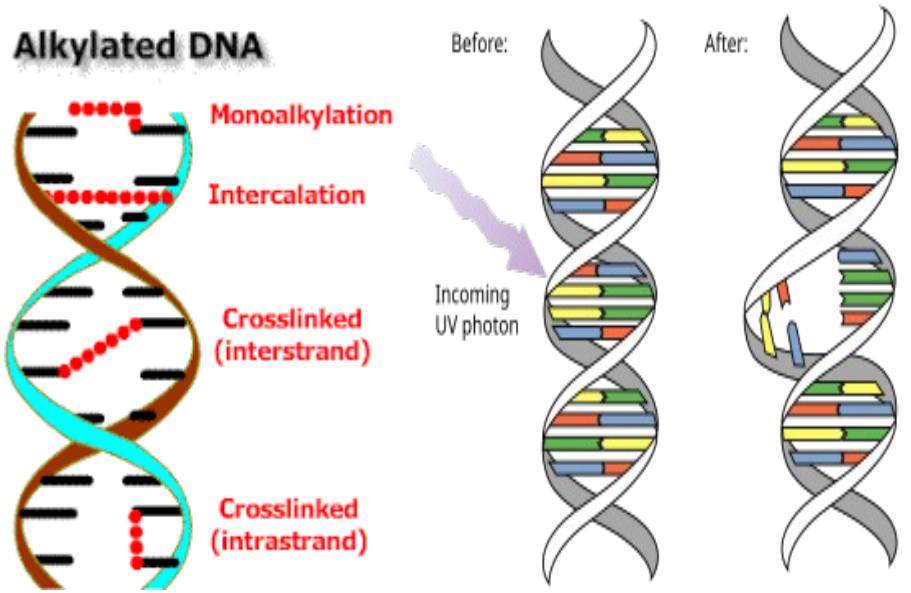
- **leukemia may be an occupational hazard for radiologists, who are routinely exposed to X-rays.**

How can radiation, such as UV light, damage DNA and trigger carcinogenesis?

Mechanism of carcinogenesis by radiation

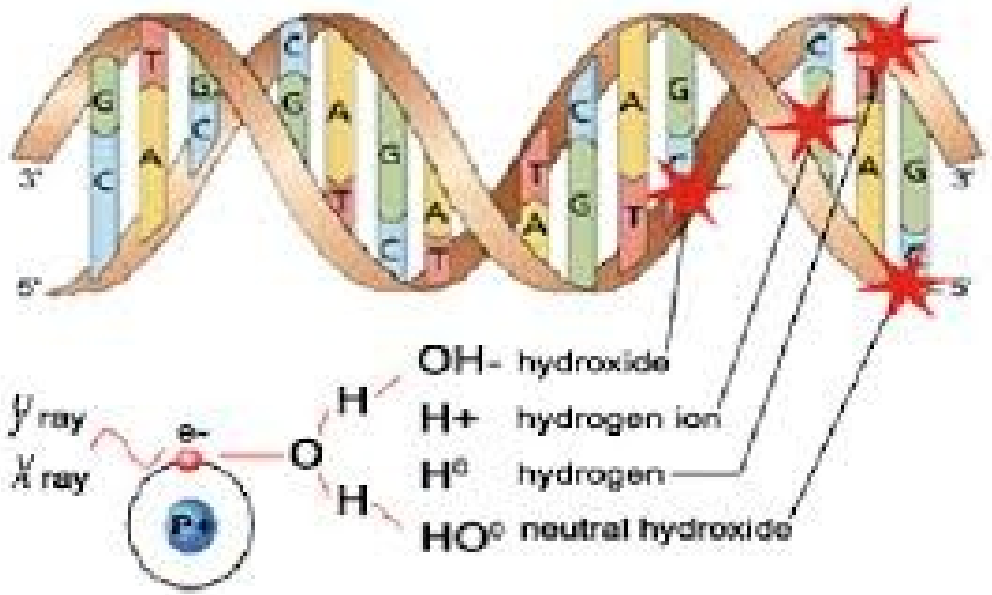
1. A direct effect like UV rays: produces damage to DNA, which may be as follows:

- **Single or double bond strands break**
- **Elimination of purine/ pyrimidine bases**
- **Formation of pyrimidine dimers**
- **Cross-linking of strands**



2- Indirect effect:

- Like X-rays & Y-rays, apart from causing direct damage to DNA,
- Ionizing radiation has the capacity to displace electrons from atoms, thus converting them to ions & generating free radicals like O^{2-}, H_2, $OH2^-$, HOO^-, OH^-, etc.
- The resultant free radicals, ther within living cell molecules.



How can chemical carcinogens trigger carcinogenesis?

2nd causes of cancer: Chemical carcinogens:

- **Several chemical substances can cause malignancy,**
- **They are collectively known as chemical carcinogens (80% of cancer is caused by chemicals).**

These chemicals may be present in

- **The foods we eat,**
- **The water we drink,**
- **The air we breathe,**
- **The clothing we wear,**
- **Sometimes in medicines we take and in**
- **The workplace we work in**

Note:

- **Chemical carcinogens require long exposure to produce their carcinogenic effects, which vary with the type & amount of the carcinogen**

How are chemical carcinogens classified?

Classification of chemical carcinogens:

According to their occurrence

I. Chemicals that are used as drugs, such as

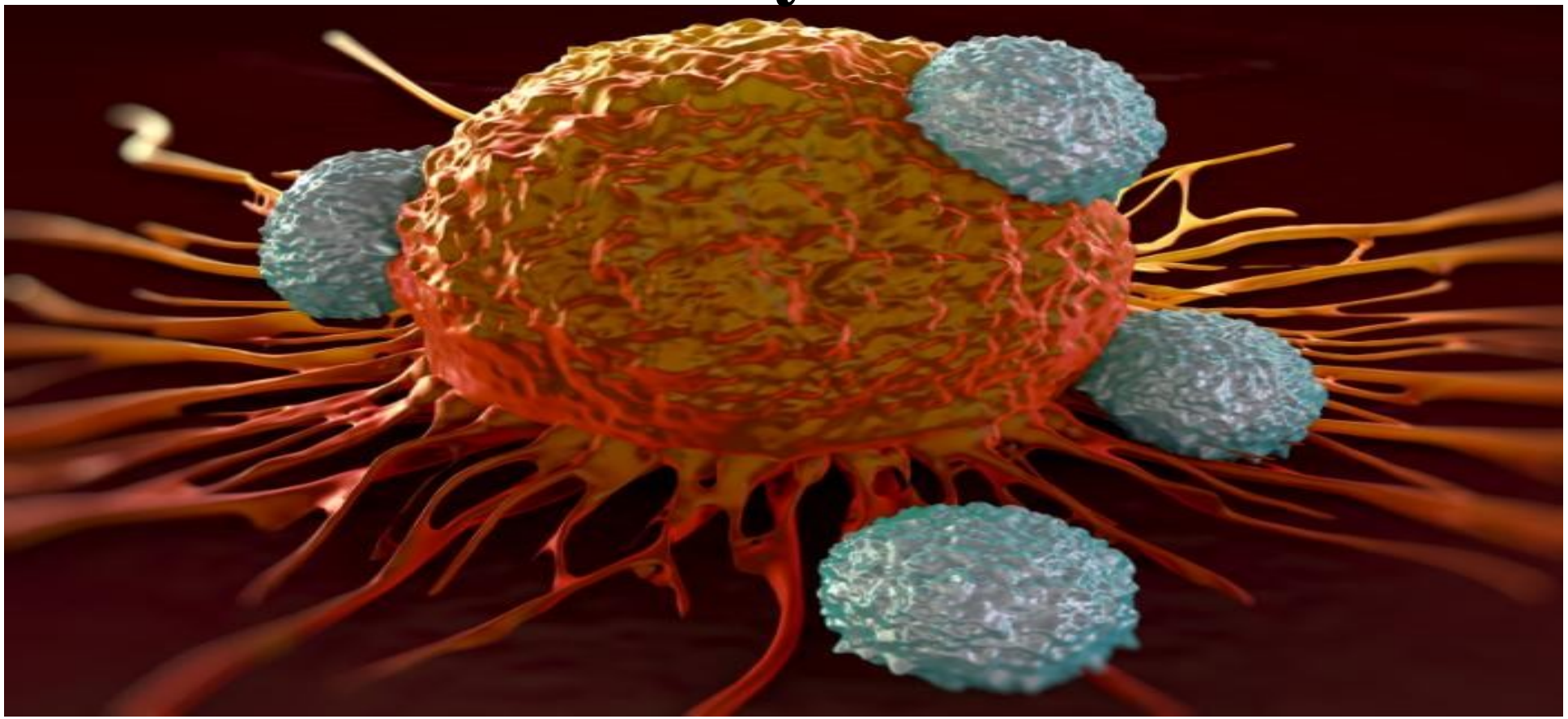
- **Exogenous hormones (Stilbesterol)**
- **Immunosuppressive drugs,**
- **Isoniazid, Cholamphenicol,**

II. Chemicals which cause occupational malignancy, such as

- **Asbestose, CCl4,**
- **Polycyclic Aromatic Hydrocarbons (PAH),**
- **Aromatic amines, Nitrosamines,**

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What is the mechanism by which chemical carcinogens lead to cancer development?

What is the mechanism by which chemical carcinogens lead to cancer development?

A- Direct-acting carcinogens

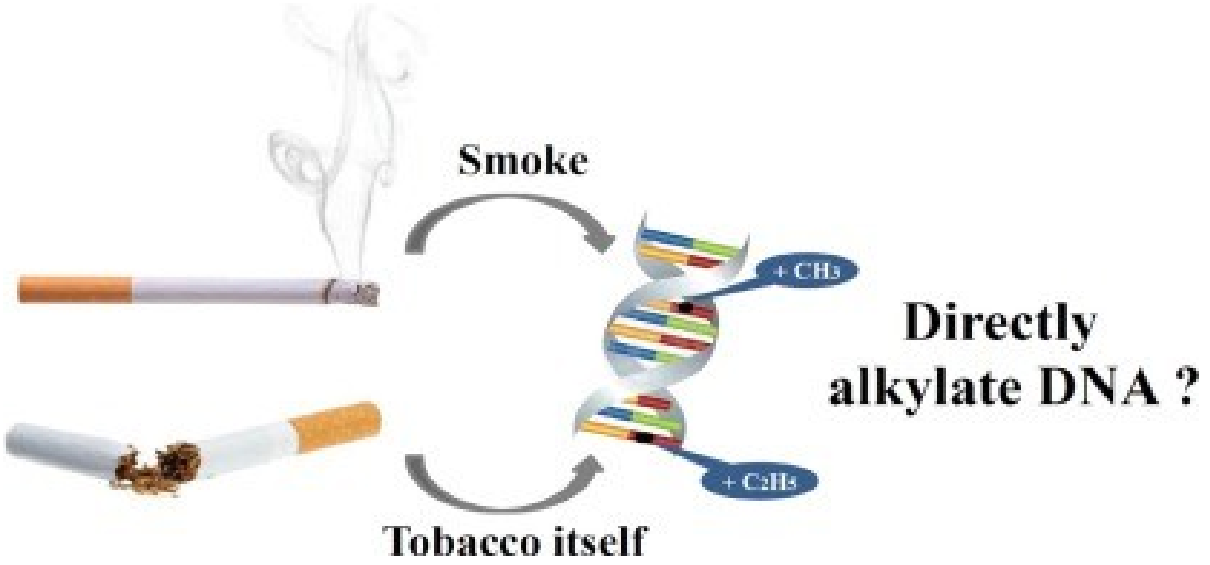
B- Indirect acting carcinogens

How do direct-acting chemicals act?

Mechanism of action;

A- Direct-acting carcinogens :

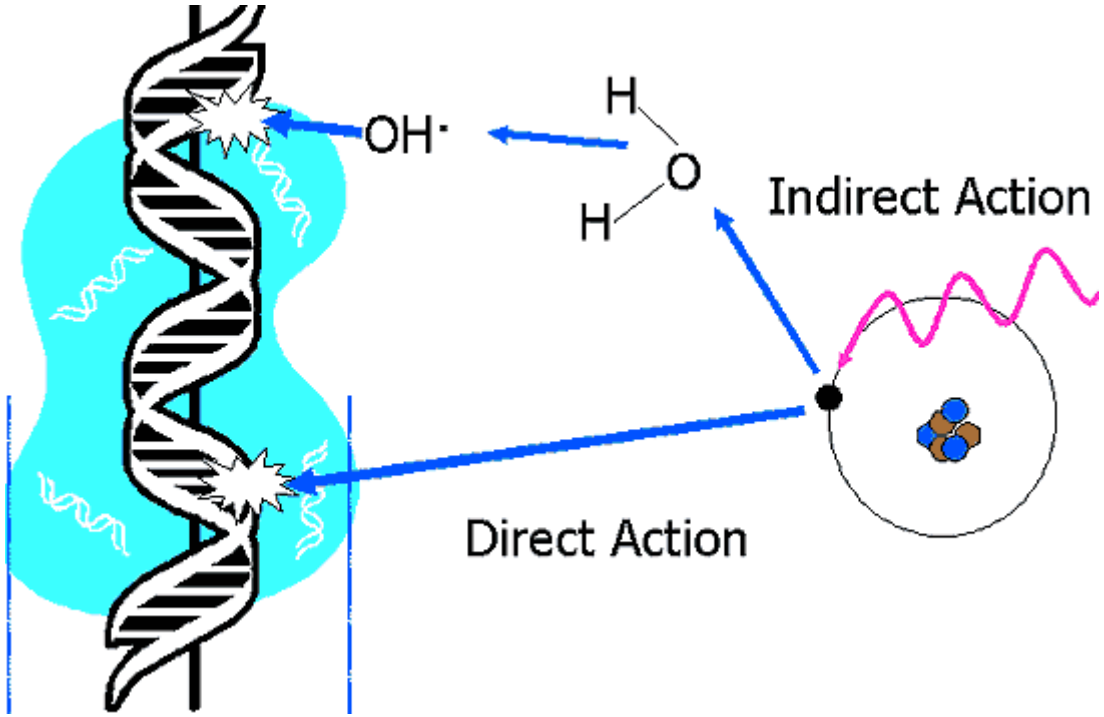
Several chemical carcinogens, such as alkylation agents (e.g., cyclophosphamide), chemotherapy drugs, and immunosuppressants. can interact directly with target molecules, including DNA, RNA, and proteins.



How do indirect-acting chemicals act?

B- Indirect acting carcinogens

- Most of the chemicals act as procarcinogens. They are not chemically reactive.
- In the body after metabolism, they are converted to ultimate carcinogens, which are highly carcinogenic thus:



Procarcinogen



Proximate(Carcinogen)



Ultimate carcinogenic
Highly carcinogenic

Note: Most of the ultimate carcinogens are electrophiles (deficient in electrons) and thus they can readily react with nucleophilic (electron-rich) groups in DNA, RNA, and various proteins

In what ways do direct and indirect genotoxic agents contribute to carcinogenesis?

Mechanism of carcinogenicity is divided into:

I. Genotoxic (DNA-reactive): they cause irreversible genetic damage or mutations.

Subdivided into:

- **Activation-independent: Direct genotoxic carcinogens such as alkylating agents (Mitomycin C)..**
- **Activation–dependent: Indirect genotoxic carcinogens such as PAH & Nitrosamines.**
- **Inorganic (Metals) such as Co, Cd, Pb, & As.**

II. None-Genotoxic (not DNA reactive):

These agents do not cause direct DNA damage but promote cell growth through other mechanisms, such as hormonal imbalance, oxidative stress, inflammation & epigenetic

These include hormones & some organic compounds.

- **Promoters:** **Saccharin**
- **Cytotoxic:** **Nitrilo triacetic acid**
- **Hormone modifying:** **Estrogen**
- **Immuno suppressor:** **Purine analogs**
- **Solid state:** **Plastics, & Asbestos**

III. Unclassified or Miscellaneous:

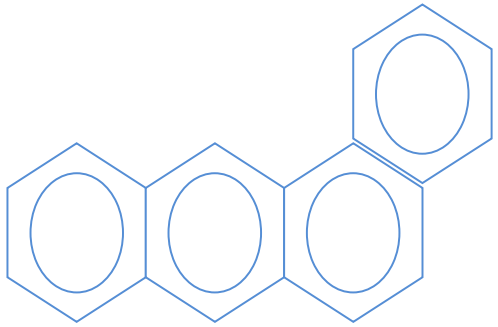
- **Dioxane**
- **Mustard gas**

Note:

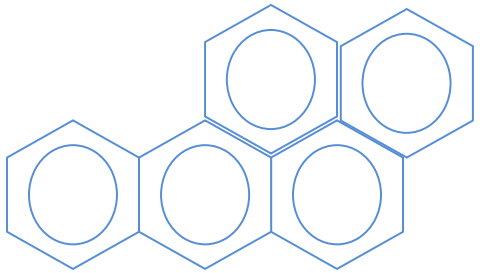
- **The majority of chemical carcinogens are activation–dependent carcinogens and require activation.**
- **activation–dependent is enzymatic, and it's a part of the detoxification process of the human body.**
- **Many cells of the body, mainly those of the liver, possess enzymes that can catalyze the conversion of pro-carcinogens to their active forms.**
-

Examples:

Polycyclic aromatic hydrocarbon (PAH)



(Benzo(a)anthracene



Benzo(a)pyrene

Mechanism of Carcinogenicity of Polycyclic Aromatic Hydrocarbons (PAHs)

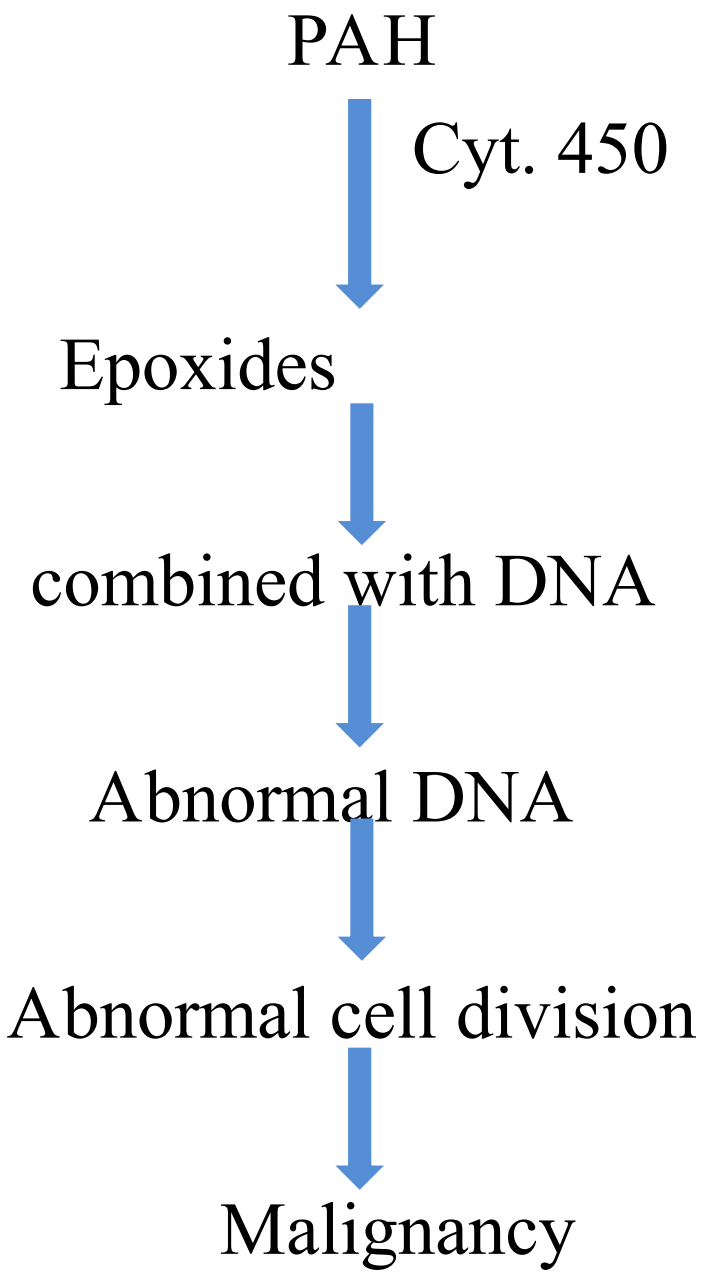
1-PAH are metabolized by the enzyme system cytochrome P450 into reactive intermediates called epoxides.

2- Some epoxides are excreted in urine, but others bind covalently to DNA, forming DNA adducts.

3- The binding of epoxides to DNA can cause mutations or structural changes in the DNA.

4-These mutations may lead to errors in cell division, disrupting normal cell growth regulation.

5- Over time, these changes can accumulate and result in uncontrolled cell proliferation, ultimately causing cancer.

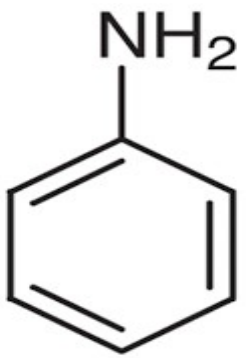


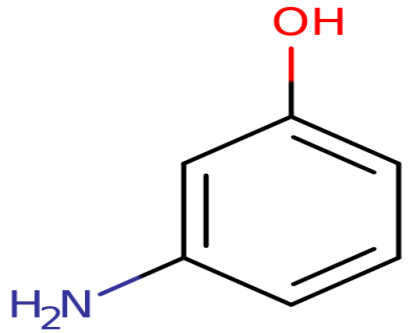
1st example of carcinogenic substance:

Aromatic amines

like: Aniline:

- **Anilines become carcinogenic after their metabolism in the liver, forming OH-amino derivatives, which may be detoxified by conjugation with glucuronic acid.**
- **In the bladder, hydrolysis of glucuronide derivatives releases the active hydroxylamines, which are carcinogenic.**
- **These carcinogens cause mainly urinary tract malignancy.**

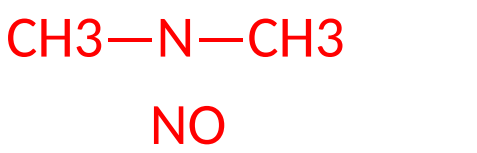




2nd example of carcinogenic substance: Nitrosamines (food preservatives)

Such as: Methyl nitrosamines

Methyl nitrosamines are activated by hydroxylation & they cause mainly malignancy of the G.I.T.



Dimethyl nitrosamine

3rd Causes of cancer- Viruses

Experimental facts proved that certain types of the virus cause certain types of cancer

- **Hepatitis B virus(is probably the major etiological agent for many liver cancers,**
- **Human papillomavirus**

4th Causes of cancer-Hormones

Certain organs of the human body respond to certain hormones by hyperplasia & hypertrophy, Such as

- **Estrogen,**
- **Testosterone &**
- **Prolactin.**

Principle of cancer treatment

What methods are being used to treat cancer today?

The 4 major approaches to cancer treatment are:

a. Surgery.

b. Radiotherapy

c. Chemotherapy

d. Biological therapy

A- Surgery

In many patients with cancer, the tumor is discovered before it has spread, and it may be cured by surgical excision. like, breast,

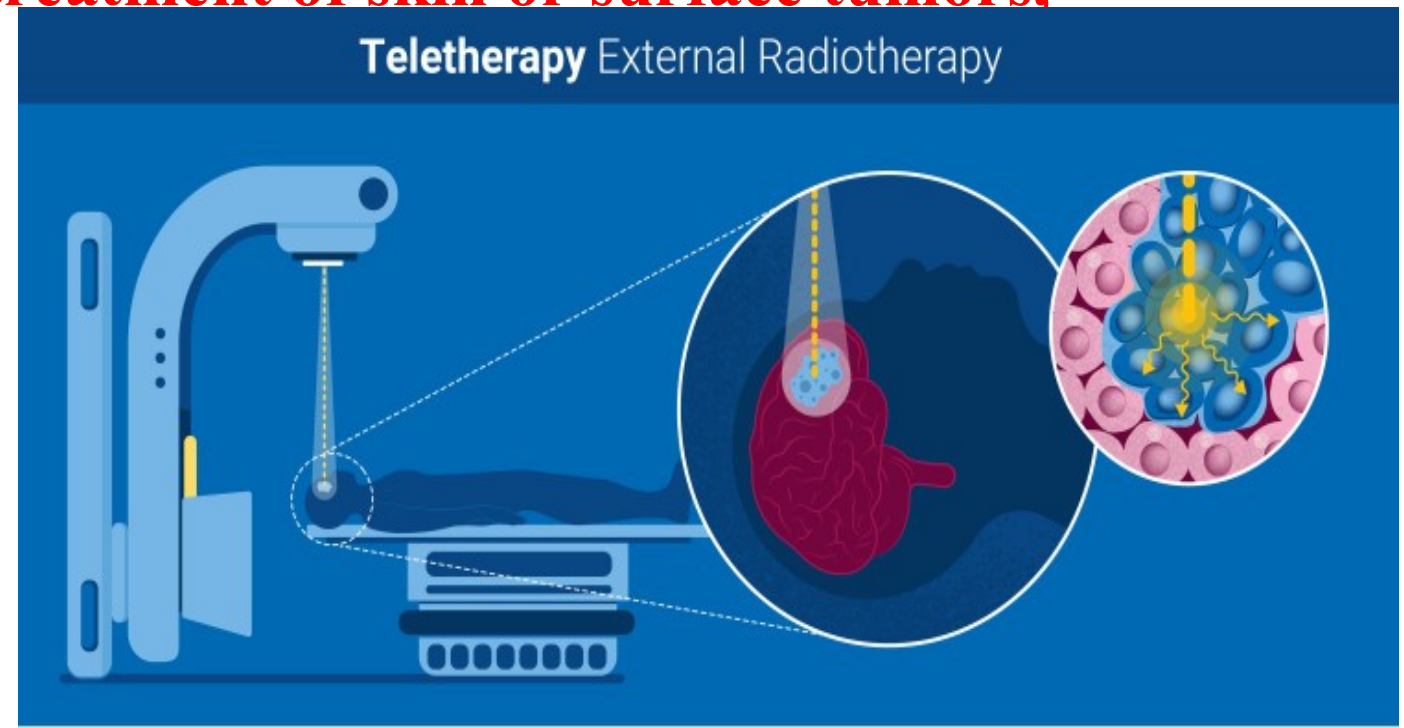
A list of terms that describe surgical procedures used in treating cancer

- **Cryosurgery: malignant tissue is frozen and thus destroyed.**
- **Electrocauterization: malignant tissue is destroyed by burning**
- **Excisional biopsy**

B- Radiotherapy:

The goal of radiation therapy is to deliver a maximal dose of ionizing radiation to the tumor tissue & a minimal dose to the surrounding normal tissue, like:

- **Electron beams,**
- **Low-energy beams for the treatment of skin or surface tumors,**
- **Proton therapy**



C- Chemotherapy

- **It is the standard treatment for many types of cancer.**
- **It may be used alone or in combination with surgery & radiation to improve cure rates**
- **A large number of drugs are used in the treatment of cancers, they are collectively known as cytotoxic drugs.**

Depending on their biochemical mechanism of action, they are classified into

1-Alkylating agents

This process may prevent or result in the imperfect division of the DNA, fragmentation of chromosomes, and so cell death may result.

Enzymes concerned with DNA synthesis may also be alkylated.

Examples:

- **Nitrogen mustard (Mustine),**
- **Cyclophosphamide (Endoxan)**

2- Antimetabolites:

- **It inhibits the synthesis of substances that are the necessary components of DNA, or it may directly block the replication of DNA**
- **Are compounds closely resemble metabolites essential to the synthesis of nucleic acids and proteins.**
- **They are therefore incorporated into natural metabolic pathways & enzyme systems.**

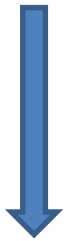
Examples:

- **Methotrexate,** **Used for: Cancers: Leukemia (especially ALL), breast cancer, osteosarcoma, lymphoma. Autoimmune diseases: Rheumatoid arthritis, psoriasis, Crohn's disease. Ectopic pregnancy: Non-surgical treatment.**
- **5-Fluorouracil** **(Colon cancer, breast cancer, gastric cancer)**
- **6-Mercaptopurine,** **used for Acute lymphoblastic leukemia (ALL & Immunosuppression in some autoimmune conditions**

Methotrexate

This is a folic acid antagonist, inhibiting the dihydrofolate reductase, which normally reduces

FH_2

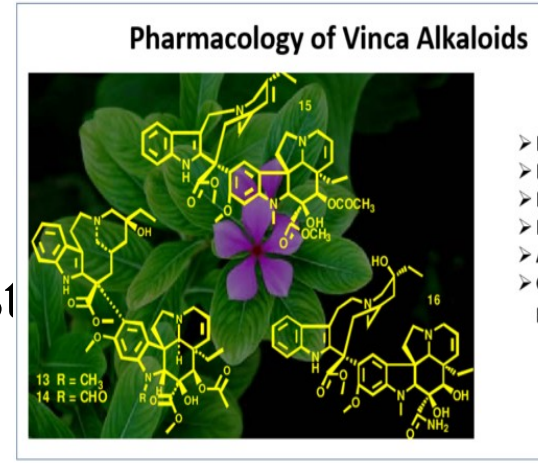


FH_4

FH4 is an essential co-enzyme in the formation of purine & pyrimidine bases, consequently, DNA synthesis & cell division are inhibited.

3-Antimitotics

Certain plant alkaloids such as Vinca alkaloid acts on mitosis mainly by blocking the function of the cell structural protein, which is essential for mitosis.



4- Antibiotics:

- **They are produced by bacteria or fungi.**
- **They act by binding to DNA in the cell. Thus promoting DNA st preventing normal replication.**
- **Many antibiotics inhibit tumor cell division, some of them form irreversible complexes with DNA molecules & so fragments DNA chain, so they interfere with DNA replication.**
- **Examples: Antimycin D & Mitomycin.**

5- Hormonal agents:

Some steroids, such as estrogens and androgens, have growth-inhibiting effects on leukemia & breast cancer

6- Enzymes such as Asparaginase

- Generally, neoplastic cells require more asparagine than normal cells
- The enzyme asparaginase hydrolyzes asparagine into aspartic acid & ammonia.
- Thus depriving the malignant cells of this acid, this process causes cell death with similar damage to normal tissues.

What are the most common side effects of chemotherapy?

Common Side Effects of Chemotherapy

- **Loss of hair (Alopecia)**
- **Myelosuppression.**
- **Nerve damage.**
- **Oral & digestive tract toxicity, including**
  - **Stomatitis,**
  - **Nausea**
  - **Vomiting, &**
  - **Gastrointestinal disturbances**

The general effects of malignancy:

The main effects of malignancy, especially when it's widespread, include:

- **Feeling of illness**
- **Anorexia**
- **Weight loss**
- **Abnormal metabolism**

What are the most common Characteristics of cancer cells?

Characteristics of cancer cells

- Morphological change
- Biochemical change

Morphological change

- Round shape, larger than normal
- Altered nuclear: cytoplasmic ratio
- Can grow without attaching to the surface (in vitro),

Biochemical change

- Increased synthesis of DNA and RNA
- Increased rate of glycolysis, both aerobic and anaerobic
- Show alteration of permeability and surface change
- Alteration of oligosaccharide chains
- Increased activity of ribonucleotide reductase and decreased catabolism of pyrimidines
- The Appearance of new antigens and the loss of certain antigens
- Change of glycolipids and glycoprotein constituents on the cell surface
- Alteration of the activities of certain enzymes, such as proteases
- Alteration in transport proteins
- Inappropriate synthesis of certain hormones and growth factors
- Electrolyte and endocrine disturbance.